Korea Technology and Information Promotion Agency for SMEs

Technology Proposal for 2026 TIPA* International Joint R&D Program
(TIPA: Korea Technology and Information Promotion Agency for SMEs)

The objective of this Technology Proposal is to introduce the institute's proprietary technologies and present potential collaborative areas for Korean SMEs, facilitating joint research and development projects under the 2026 TIPA International Joint R&D Program. This Technology Proposal will be publicly disclosed in conjunction with the official announcement of the 2026 TIPA International Joint R&D Program.

1. Technology Proposer Information

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2. Technology Overview

Title of Proposed Technology	A Memory-Efficient Inference Infrastructure for Large AI Models			
Korea's 12 National Strategic Technology Fields	Semiconductor/Display ( ), Secondary Battery ( ), Advanced Mobility ( ), Next-gen Nuclear Power ( ), Advanced Bio ( ), Aerospace/Marine ( ), Hydrogen ( ), Cybersecurity ( ), AI (√), Next-gen Communication ( ), Advanced Robotics & Manufacturing ( ), Quantum ( )			
Industrial Technology Classification [1]	**Technology Field 1** (Large Category)		**Technology Field 2** (Large Category)	
	Information & Communication			
	Medium Category	Small Category	Medium Category	Small Category
	Software	Embedded Software		
Concept of Proposed Technology	The proposed technology introduces an adaptive memory management infrastructure for large AI models. Its main functions are to reduce inference cost, improve scalability, and enable efficient processing of long and evolving inputs on existing hardware platforms. The technology can be used in AI-powered products such as virtual assistants, analytics tools, and decision-support systems deployed by small and medium-sized companies.			
Rationale for Joint R&D Project	• Small and medium-sized companies face significant cost and scalability barriers when deploying large AI models in real-world products. • Existing AI efficiency solutions are often difficult to integrate into production systems and require specialized expertise that is not readily available in smaller organizations. • There is a strong need for practical, deployment-ready AI infrastructure that reduces inference cost while supporting advanced, long-context, and interactive AI applications. • Industry partners provide real-world use cases, operational constraints, and validation environments that are essential for shaping practical system requirements and evaluating performance under realistic conditions.			

	• The research institute contributes specialized expertise in AI efficiency, memory-aware inference, and system-level optimization that is typically unavailable within SMEs. • Joint R&D enables rapid iteration, real-world testing, and efficient transition from prototype to commercial deployment, ensuring that the developed technology meets both technical and market requirements.
Key Words	1. AI Inference efficiency 2. Memory optimization 3. Scalable AI infrastructure 4. Cost-efficient deployment

3. Research and Development Objectives & Plan

Technology Readiness Level (TRL)[2]	**Start of the Project**	Level 2	**End of the Project**	Level 4
R&D Objectives	• Develop an adaptive AI memory management technology that reduces runtime memory usage and inference cost for large AI models. • Create a deployable AI infrastructure software layer that integrates with existing models and inference pipelines without requiring retraining or changes to application logic. • Enable efficient long-context and multi-step AI inference on cost-constrained hardware commonly used by small and medium-sized companies. • Design and implement hardware-compatible memory optimization modules that improve performance on modern GPUs and AI accelerators. • Deliver and validate industry-ready prototypes and pilot systems to ensure commercialization readiness within 3 years.			
R&D Target Specifications	• Runtime Memory Reduction ○ Achieve 40–70% reduction in AI inference memory usage compared to standard deployment without memory optimization. ○ Enable support for 2x–4x longer input contexts on the same hardware. • Inference Performance ○ Improve end-to-end inference throughput by 20–50% for long-context workloads. ○ Limit per-token latency overhead introduced by the technology to ≤5–10%. • Accuracy Preservation ○ Maintain model task performance within ≤2–3% degradation compared to unoptimized inference across representative workloads. • Hardware and Platform Compatibility ○ Demonstrate compatibility with commercial GPUs and AI accelerators without requiring hardware modification. ○ Support deployment on cloud, on-premise, and hybrid environments commonly used by small and medium-sized companies. • Deployment Readiness ○ Deliver a production-ready software prototype that can be integrated into existing AI inference pipelines within a short deployment cycle. ○ Validate the technology through at least one industry-relevant pilot deployment by the end of the project.			
R&D Plan	**Overall Approach and Methodology** The goal of this joint R&D project is to develop a practical, deployable AI infrastructure technology that enables small and medium-sized companies to run large AI models more efficiently in real-world settings. The project follows a 3-year, application-driven R&D approach that combines applied research, system engineering, industry integration, and commercialization-oriented development. The overall methodology emphasizes iterative system design, prototype-driven development, pilot-			

based validation, and progressive technology maturation. Rather than pursuing fundamental algorithmic research, the project adopts an engineering-first methodology focused on translating prior research insights into a robust, scalable, and industry-ready infrastructure solution.

Existing approaches to improving AI inference efficiency typically rely on fixed compression, quantization, or heuristic memory pruning strategies [1-7]. While effective in controlled settings, these methods are often static, difficult to adapt to evolving workloads, or tightly coupled to specific model architectures [8, 9]. As a result, they provide limited flexibility for long-context, multi-step, or interactive AI applications commonly encountered in industrial systems.

The proposed technology builds on the research team's prior work in memory-efficient inference for large language models [10-12]. In earlier projects, the team developed sentence-level memory management techniques [10] that reduce redundancy in long-context inference, as well as multi-granularity memory access mechanisms [11] that enable models to selectively expand or compress internal representations during reasoning. These approaches demonstrated that substantial memory and latency reductions are achievable without sacrificing model performance, providing a strong technical foundation for the proposed infrastructure. Building on this experience, the current project consolidates and generalizes these ideas into a unified, production-oriented AI memory infrastructure designed explicitly for industrial deployment. Figure 1 illustrates the high-level integration of the proposed memory optimization layer within a standard AI inference pipeline, highlighting how the technology augments existing systems without disrupting application logic.

The R&D plan is organized into three phases aligned with increasing technology maturity: (1) core technology development, (2) industry integration and validation, and (3) commercialization readiness.

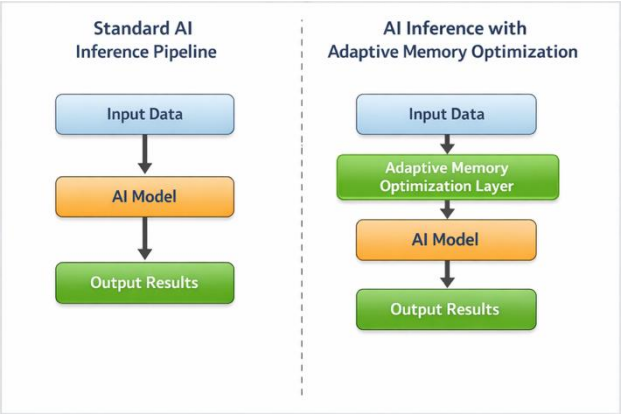

Figure 1. High-level system overview illustrating a standard AI inference pipeline and the same pipeline augmented with the proposed adaptive memory optimization layer.

Year 1: Core Technology Development and Prototyping

Research Approach and Key Tasks

The first year focuses on establishing the foundational software infrastructure for adaptive AI memory management and delivering an initial working prototype. The primary research approach is prototype-driven system development, in which core design concepts are rapidly implemented, evaluated, and refined.

Key tasks include designing the system architecture, implementing the core components of the memory optimization layer, and defining modular interfaces compatible with widely used AI frameworks and deployment environments. Initial performance evaluations will be conducted on representative long-context AI workloads to quantify memory reduction, runtime efficiency, and integration overhead. These evaluations will serve as internal benchmarks guiding subsequent

refinement.

Expected Milestones

By the end of Year 1, the project will deliver:

- A functional prototype of the memory optimization infrastructure
- Initial performance benchmarks demonstrating measurable efficiency gains
- Basic documentation and integration guidelines for early adopters

Year 2: Industry Integration and Validation

Research Approach and Key Tasks

The second year centers on pilot-driven validation and system refinement through close collaboration with industry partners. The methodology emphasizes application-guided development, where real-world use cases shape system configuration, performance targets, and usability requirements.

Key tasks include integrating the technology into partner applications, configuring deployment environments, and conducting pilot deployments under realistic operating conditions. Performance will be evaluated in terms of memory usage, inference cost, latency, scalability, and operational robustness. Figure 2 shows a representative deployment workflow illustrating how industry partners integrate the proposed technology into existing AI applications and production pipelines. Feedback from these deployments will guide iterative improvements to robustness, configurability, and usability.

Expected Milestones

By the end of Year 2, the project will deliver:

- Validated pilot deployments in industry-relevant applications
- Quantitative performance reports demonstrating real-world efficiency gains
- Refined system components ready for production hardening

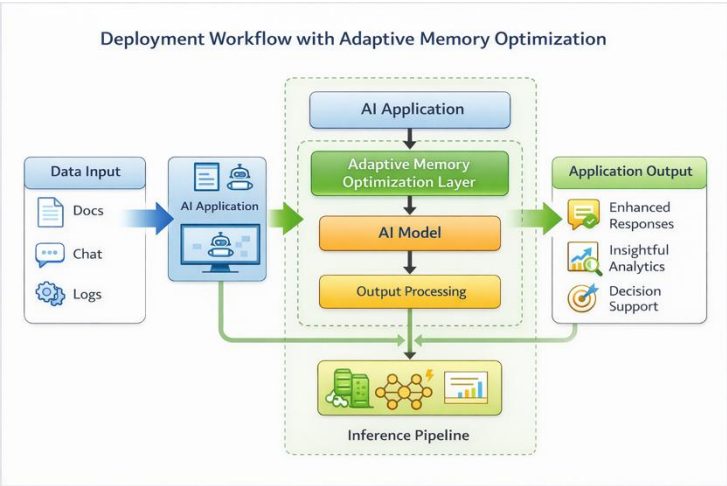

Figure 2. Deployment workflow showing how industry partners integrate the proposed technology into existing AI applications, from data input through inference to application output.

Year 3: Commercialization Readiness and Scaling

Research Approach and Key Tasks

The third year focuses on engineering maturation and commercialization preparation. The methodology emphasizes system hardening, scalability testing, and usability engineering to ensure

production readiness. Key tasks include improving system stability, optimizing deployment workflows, developing user-facing configuration and monitoring tools, and expanding documentation to support adoption by non-expert users. In parallel, the team will work with industry partners to define commercialization pathways, including licensing models, product integration strategies, and potential startup spin-off opportunities.

Expected Milestones

By the end of Year 3, the project will deliver:

- A production-ready AI memory optimization infrastructure
- Deployment-ready software packages and documentation
- A defined commercialization and technology transfer plan

Figure 3 summarizes the overall development timeline and key milestones across the three project phases.

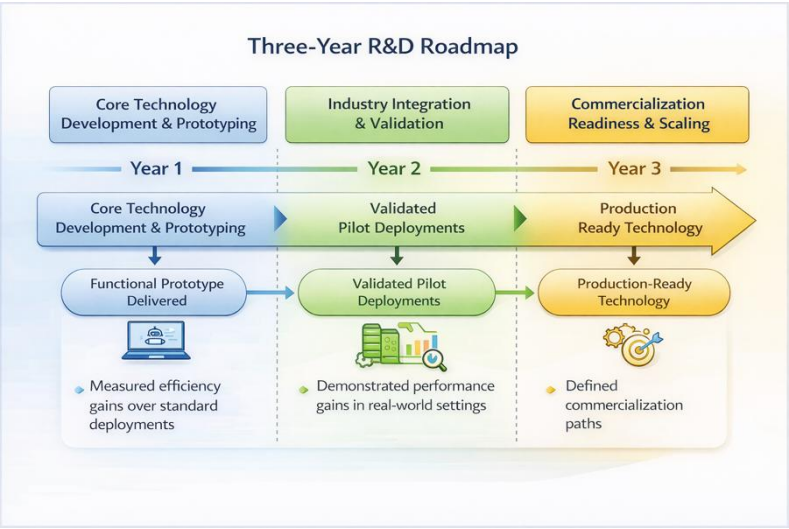

Figure 3. Three-year roadmap illustrating key milestones, deliverables, and increasing technology readiness from prototype to commercial deployment.

Expected Outcomes and Impact

At the conclusion of the project, the joint R&D effort will deliver a deployable AI memory optimization technology validated through industry collaboration. The technology will enable companies to run larger and more capable AI models at lower cost, improve scalability for long-context applications, and reduce dependence on expensive hardware. These outcomes directly support the needs of startups and small-to-medium enterprises seeking to integrate advanced AI into commercial products within a short timeframe.

This plan emphasizes engineering deliverables, pilot validation, and commercialization readiness. It leverages prior research experience to reduce technical risk while focusing on practical impact, making it well aligned with the goals of industry-driven, short-horizon R&D programs.

4. R&R (Roles and Responsibilities)

Expected R&R of Research Institute	The research institute will serve as the technical lead and primary developer of the proposed AI memory optimization technology. Its role is to provide the core scientific, engineering, and system-level expertise required to design, implement, and validate a deployable AI infrastructure solution suitable for industrial adoption. **Technical Leadership and Core Technology Development** The research institute will lead the overall technical design and architecture of the proposed memory optimization infrastructure. Drawing on its prior work in AI efficiency, memory-aware inference, and scalable AI systems, the institute will define the system framework, develop the core software components, and establish performance benchmarks. This includes designing modular interfaces compatible with widely used AI frameworks and ensuring that the technology integrates seamlessly into existing inference pipelines without requiring changes to application logic, deployment workflows, or model retraining. **Research Scope and Methodology** The institute will be responsible for conducting the applied research necessary to translate prior algorithmic insights into a production-oriented infrastructure solution. This includes exploring adaptive memory management strategies, evaluating trade-offs between efficiency and accuracy, and identifying robust configurations suitable for long-context and interactive AI workloads. The research scope will remain focused on engineering feasibility and deployment readiness, rather than fundamental algorithmic research. **Prototype Implementation and System Integration** The institute will develop and maintain the core prototype implementations of the proposed technology. It will provide reference implementations, testing frameworks, and integration guidelines to support deployment in industry partner environments. The institute will also support the integration of the technology into pilot applications, working closely with industry collaborators to ensure compatibility with real-world deployment constraints. **Performance Evaluation and Validation** The institute will lead the design and execution of performance evaluations, including benchmarking memory usage, inference latency, throughput, and accuracy preservation under representative workloads. These evaluations will be used to guide iterative system refinement and to document measurable efficiency gains required for commercialization readiness. **Resource Provision and Infrastructure Support** The research institute will provide the necessary computational resources, software infrastructure, and research personnel to carry out the project. This includes access to GPU-based compute platforms, development environments, and testing frameworks required for large-model inference experiments. Faculty researchers, graduate students, and dedicated research engineers will contribute to system development, testing, and documentation. **Knowledge Transfer and Commercialization Support** The institute will play a central role in knowledge transfer and technology maturation, preparing technical documentation, deployment guides, and training materials to facilitate adoption by industry partners. It will collaborate with partners to define commercialization pathways, support technology licensing or integration, and contribute to potential startup formation or productization efforts.

Expected R&R of Korean SME Partner	The Korean SME partner will serve as the primary industrial collaborator and commercialization driver for the proposed joint R&D project. Its role is to provide real-world application contexts, deployment environments, and product integration pathways that ensure the developed technology addresses practical industry needs and can transition efficiently into commercial products. **Application Definition and Use-Case Leadership** The SME partner will identify and define industry-relevant use cases where efficient deployment of large AI models is critical, such as intelligent document processing, interactive assistants, analytics platforms, or decision-support systems. These use cases will guide system requirements, performance targets, and integration constraints throughout the project. The partner will provide representative workloads, application scenarios, and operational requirements that shape the design of the proposed technology. **System Integration and Product Development** The SME partner will lead the integration of the proposed technology into its existing AI applications and product pipelines. This includes adapting application interfaces, configuring deployment environments, and embedding the memory optimization layer into production inference workflows. The partner will contribute engineering resources to ensure compatibility with its software stack, cloud or on-premise infrastructure, and operational monitoring systems. **Testing, Validation, and Performance Assessment** The SME partner will play a central role in testing and validating the technology under real-world operating conditions. This includes conducting pilot deployments, collecting performance metrics, and evaluating memory usage, latency, throughput, reliability, and user experience. The partner will provide feedback on system behavior, usability, and operational robustness, enabling iterative refinement toward production readiness. **Resource Commitment and Infrastructure Support** The SME partner will provide the necessary development personnel, application environments, and deployment infrastructure required for pilot testing and validation. This includes access to representative datasets, production-like inference systems, and monitoring tools used in operational settings. Dedicated engineers and product managers will participate in joint development, integration, and evaluation activities. **Commercialization and Market Deployment** The SME partner will take a leading role in commercialization planning and market deployment. This includes identifying target customer segments, defining product offerings incorporating the developed technology, and planning go-to-market strategies. The partner will collaborate with the research institute to explore licensing, product integration, and potential spin-off opportunities, ensuring that the technology transitions from prototype to revenue-generating products within the project timeframe. **Collaboration and Project Coordination** The SME partner will participate actively in joint project planning, regular technical meetings, and milestone reviews. It will coordinate closely with the research institute to align technical development with product timelines and business objectives. This close collaboration ensures rapid feedback, risk mitigation, and efficient progress toward commercialization.

5. Expected Outcomes/Impact

Expected Outcomes and Impact	**Technological Outcomes** The primary technological outcome of this joint R&D project will be a deployable AI memory optimization infrastructure that significantly improves the efficiency, scalability, and cost-effectiveness of large AI model inference. The project will deliver a production-ready software system capable of reducing runtime memory usage, lowering inference cost, and enabling long-context and multi-step AI applications on existing hardware platforms. Key technical deliverables include a modular memory optimization layer compatible with widely used AI frameworks, reference implementations integrated into industry pilot systems, and validated performance benchmarks demonstrating measurable gains in memory efficiency, throughput, and deployment scalability. These outcomes establish a practical foundation for advanced AI applications that would otherwise remain too costly or resource-intensive for small and medium-sized companies. **Commercialization Outcomes** A central outcome of the project is the transition of the developed technology from prototype to commercial deployment. Through close collaboration with the Korean SME partner, the project will deliver validated pilot applications and a production-ready system suitable for integration into commercial AI products and services. The expected commercialization results include the incorporation of the technology into the SME partner's product portfolio, the development of market-ready AI solutions with improved performance and lower operational cost, and the establishment of licensing or product integration pathways. These outcomes create direct opportunities for revenue generation, product differentiation, and expansion into new customer segments that require scalable and cost-efficient AI capabilities. The project also creates a foundation for future commercialization channels, including technology licensing to additional companies, adoption by other SMEs, and potential startup or spin-off activities built around efficient AI infrastructure. **Industrial Impact** The project addresses a critical barrier to AI adoption in industry: the high cost and limited scalability of large-model deployment. By lowering infrastructure requirements and enabling efficient long-context inference, the technology expands access to advanced AI capabilities for startups and small-to-medium enterprises that lack large AI engineering teams or high-end hardware resources. The expected industrial impact includes: • Enabling broader deployment of AI in document processing, customer support, analytics, and decision-support systems. • Improving competitiveness of SMEs by allowing them to offer more capable AI-powered products at lower cost. • Accelerating adoption of AI technologies across software, service, and manufacturing sectors where long-context and interactive AI applications are increasingly important. These impacts contribute to a more inclusive and scalable AI ecosystem, reducing concentration of advanced AI capabilities among only large technology firms. **Economic and Ecosystem Impact**

	The project contributes to economic growth and innovation by strengthening the AI infrastructure capabilities of Korean SMEs and supporting the development of high-value AI products and services. By reducing deployment cost and improving scalability, the technology lowers entry barriers for new AI-driven businesses and supports job creation in AI engineering, product development, and digital services. At the ecosystem level, the project fosters a sustainable academia–industry collaboration model that accelerates technology transfer from research to market. The outcomes strengthen national competitiveness in AI infrastructure technologies, support the growth of domestic AI companies, and contribute to the development of a robust AI innovation ecosystem with global market relevance. **Long-Term Impact** Beyond the project horizon, the developed technology establishes a long-term platform for efficient AI deployment applicable to future generations of large and multimodal models. The infrastructure can be extended to new applications, hardware platforms, and industry domains, ensuring lasting relevance as AI systems continue to grow in scale and complexity. Overall, the joint R&D project delivers not only a commercially viable technology but also a strategic contribution to scalable AI adoption, enabling sustained industrial innovation and economic impact.

6. References

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☐ Terms:

1) Industrial Technology Classification

· Classification system according to Industrial Technology Classification Table (**Appendix 1**, based on Article 16 of the Common Operational Guidelines for the Industrial Technology Innovation Project in Korea).

2) Technology Readiness Level (TRL)

Description		
·A consistent and objective indicator of the technological maturity of key element technologies		
	Basic Research Stage (1-2)	1st Stage: Basic theories and experiments 2nd Stage: Proof of Concept, such as practical ideas or patents
	Experimental Stage (3-4)	3rd Stage: Verification of basic performance at a laboratory scale 4th Stage: Evaluation of key performance of materials, components, and systems at a laboratory scale
	Prototype Stage (5-6)	5th Stage: Production and performance evaluation of prototypes of finalized materials, components, and systems 6th Stage: Production and performance evaluation of prototypes at a pilot scale
	Productization Stage (7-8)	7th Stage: Reliability testing and evaluation by potential customers 8th Stage: Prototype certification and standardization
	Commercialization Stage (9)	9th Stage: Commercialization